PO12Q - Introduction to Quantitative Political Analysis II

Mock Exam 2025/26

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Paper Code: PO12Q –
Exam Period: Mock Exam

Exam Rubric

Time Allowed: 84 Minutes
Exam Type: Written - Standard
Approved Calculators: Permitted

Answer all questions in the answer booklet provided

UNIVERSITY OF WARWICK

Table of Contents

Instructions

- Part I contains the questions for the exam
  - These are divided into four blocks: A, B, C, and D.
  - Block A is compulsory and must be answered by all candidates.
  - You must then select ONE block out of B, C, and D.
  - Should you change your mind during the exam, indicate clearly which block you wish to have marked.
  - You can achieve a maximum of 84 points.
  - Information on the maximum number of points obtainable is outlined at the beginning of each block
- Part II contains the case study to which all questions in Block A of Part I refer
- Part III contains the relevant statistical tables
  - Normal curve tail probabilities are provided in Table 5, page 12.
  - t-distribution critical values are provided in Table 6, page 13.
  - χ^2-distribution critical values are provided in Table 7, page 14.
- Part IV contains the formula collection and notation of the module
  - If you use notation other than in the formula collection (page 15) or notation table (page 17), give an explanation.

Part I: Exam Questions

1 | Block A

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60 points, points for each question in parentheses

All questions in this section relate to the case study in Part II.

Level 1 (each question 4 points)

1. Subsections 13.1 and 13.2: which model building strategy has been employed here? Explain your answer.
2. Subsections 13.1 and 13.2: State the null- and alternative hypotheses underpinning Model 2.

Level 2 (each question 6 points)

3. Interpret the output of Section 11.
4. Subsection 12.1: Can we conclude that the proportion of democracies differs between low- and high-GDP countries in sub-Saharan Africa? Show your reasoning with reference to the output.
5. Table 4: Interpret the intercept in Model 1.
6. Table 4: Interpret the coefficient of "Primary Sector, value added (% of GDP)" in Model 5.
7. Table 4: Interpret the model fit for Model 5.
8. We have applied a logarithmic transformation to per capita GDP. Would a similar transformation make sense for life expectancy? Why / why not?

Level 3 (each question 8 points)

9. Interpret the output in Subsection 14.1. Which next steps do you need to take to address the issue if applicable?
10. Given the output in Subsection 15: Is the effect size in Model 1 in Table 4 large enough for conventional levels of statistical power? Explain your reasoning and what this means substantively.

2 | Block B

24 points, 6 points for each statement

State whether each of the following statements is true or false. Explain why the statement is true or false.

1. The method of Ordinary Least Squares (OLS) minimises the sum of the distances between the observations and the regression line.
2. It is more sensible to run one multiple regression model with z independent variables than z bivariate regression models.
3. R^2 and Adjusted R^2 ($\bar{R}^2$) cannot be compared.
4. If the values of a variable are highly skewed, this variable cannot serve as an independent variable in an OLS regression.

3 | Block C

24 points, points for each question in parentheses

Table 1 shows the size of the lower chamber of the legislature and the population size of 5 European countries.

1. Specify the null and alternative hypothesis for the relationship between the size of the lower chamber of the legislature and the population size of a country. **(4)**
2. Fit a linear regression model of the type $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ to the data in Table 1. Specify the full sample regression function. **(20)**

Country	Size of legislature	Population in millions
Lithuania	141	2.89
Poland	460	36.5
Austria	183	9.2
Ireland	160	5.42
Germany	736	83.58

Table 1: Regression Data

4 | Block D

24 points, 8 points for each method / statement

Explain each of the following methods or concepts in your own words, and illustrate your explanation with a worked example of your choice.

1. Variance Inflation Factor
2. Homoscedasticity
3. Robust standard errors

Part II: Case Study

5 | Preface

The data set is composed of three different sources joined together:

- World Development Indicators: The World Development Indicators is a compilation of relevant, high-quality, and internationally comparable statistics about global development and the fight against poverty. The database contains 1,400 time series indicators for 217 economies and more than 40 country groups, with data for many indicators going back more than 50 years. (The World Bank, n.d.)
- BMR: This data set provides a dichotomous coding of democracy for every country in the world (222 total) from 1800 to 2020. (Miller et al., 2022)

The data in the data set is for the year 2000.

6 | Code Book

variable	label
`country`	Country Name
`democracy`	0 = Autocracy, 1 = Democracy (Miller et al., 2022)
`gdppc`	GDP per capita (current US Dollars)
`enrol_net`	School enrollment, primary (% net)
`life`	Life expectancy at birth, total (years)
`agri`	Agriculture, forestry, and fishing, value added (% of GDP)

Table 2: Variables in the `ssa` data set for 2000

7 | Preliminaries

```
wdi <- read.csv("ssa.csv")
```

8 | Descriptives for `ssa` Data Frame

	N	Mean	SD	Min	P25	Median	P75	Max
democracy	46	0.30	0.47	0.00	0.00	0.00	1.00	1.00
gdppc	46	977.44	1447.93	134.54	300.03	465.10	730.35	8063.66
enrol_net	30	64.01	18.96	26.83	53.04	63.63	82.17	99.05
life	46	53.80	6.51	44.83	48.88	52.68	57.44	72.78
agri	40	23.12	15.20	2.61	12.80	22.45	29.97	76.07

Table 3: Descriptive Statistics for 'ssa' Data Frame

9 | Proportion of missing values per variable

```
##   country democracy     gdppc enrol_net      life      agri
## 0.0000000 0.0000000 0.0000000 0.3478261 0.0000000 0.1304348
```

10 | Recoding

10.1 per capita GDP

```
ssa <- ssa %>%
  mutate(gdpcat=
           ordered(
             cut(gdppc, breaks=c(0,350,8064),
                 labels=c("low","high"))))

ssa$log.gdppc <- log(ssa$gdppc)
```

10.2 Life expectancy at Birth

```
ssa <- ssa %>%
  mutate(lifecat=
           ordered(
             cut(life, breaks=c(0,mean(ssa$life),80),
                 labels=c("low","high"))))
```

10.3 Democracy

```
ssa$democracy <- factor(ssa$democracy, labels=c("Autocracy", "Democracy"))
```

11 | Crosstabulations

```
table1 <- with(ssa, table(lifecat, gdpcat))
table1
##         gdpcat
## lifecat low high
##     low   13   14
##     high   3   16

Xsq1 <- chisq.test(table1, correct=FALSE)
Xsq1
##
##  Pearson's Chi-squared test
##
## data:  table1
## X-squared = 5.1477, df = 1, p-value = 0.02328
```

12 | Two-Sample Tests

12.1 Two-Sample Proportion Test

```
table2 <- with(ssa, table(democracy, gdpcat))
table2
##             gdpcat
## democracy   low high
##    Autocracy    9   23
##    Democracy    7    7

prop.test(table2, correct = FALSE)
##
##  2-sample test for equality of proportions without continuity correction
##
## data:  table2
## X-squared = 2.0544, df = 1, p-value = 0.1518
## alternative hypothesis: two.sided
## 95 percent confidence interval:
##  -0.52348678  0.08598678
## sample estimates:
##  prop 1  prop 2
## 0.28125 0.50000
```

13 | Regression

13.1 Regression Models

```
model1 <- lm(log.gdppc ~ life, data=ssa)

model2 <- lm(log.gdppc ~ enrol_net, data=ssa)

model3 <- lm(log.gdppc ~ agri, data=ssa)

model4 <- lm(log.gdppc ~ life + enrol_net, data=ssa)

model5 <- lm(log.gdppc ~ life + agri, data=ssa)

model6 <- lm(log.gdppc ~ enrol_net + agri, data=ssa)

model7 <- lm(log.gdppc ~ life + enrol_net + agri, data=ssa)
```

13.2 Results Table

	Dependent Variable: log(per capita GDP)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Life expectancy at birth	0.089***			0.060**	0.064***		0.049**
	(0.017)			(0.020)	(0.015)		(0.015)
% Primary School Enrolment (net)		0.034***		0.025**		0.019*	0.010
		(0.007)		(0.007)		(0.008)	(0.008)
Primary Sector, value added			−0.047***		−0.037***	−0.049***	−0.047***
			(0.007)		(0.006)	(0.013)	(0.011)
(Intercept)	1.526	4.229***	7.418***	1.545	3.743***	6.180***	4.015***
	(0.936)	(0.499)	(0.201)	(1.017)	(0.867)	(0.765)	(0.949)
Num.Obs.	46	30	40	30	40	28	28
R2	0.378	0.429	0.524	0.567	0.683	0.702	0.789
R2 Adj.	0.363	0.408	0.512	0.535	0.666	0.678	0.763

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 4: Regression Models

13.3 Regression Graphs

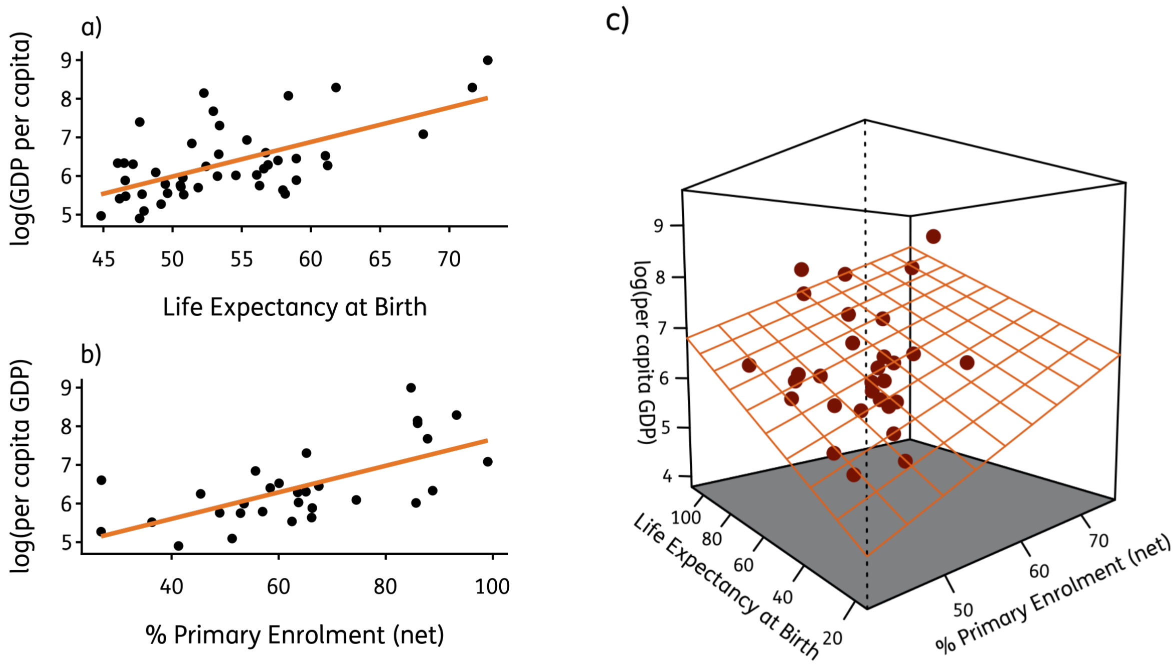


Figure 1: Visualisation of Regression Models

14 | Testing the Classical Linear Assumptions

14.1 bptest()

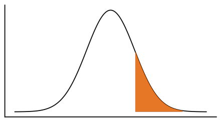
```
bptest(model1)
##
##  studentized Breusch-Pagan test
##
## data:  model1
## BP = 0.29415, df = 1, p-value = 0.5876
```

```
library(pwr)

R2 <- 0.16
f2 <- R2 / (1 - R2)
pwr.f2.test(u = 1,
            v = 46 - 1 - 1,
            f2 = f2,
            sig.level = 0.05)
##
##      Multiple regression power calculation
##
##               u = 1
##               v = 44
##              f2 = 0.1904762
##       sig.level = 0.05
##           power = 0.8250836
```

Part III: Statistical Tables

16 | Normal-Distribution



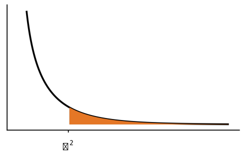
z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641
0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002

Table 5: Right-Tail Probabilities under the Normal Distribution

17 | t-Distribution

	Confidence Level									
	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	One-Tail Probability									
df	$t_{0.25}$	$t_{0.20}$	$t_{0.15}$	$t_{0.10}$	$t_{0.05}$	$t_{0.025}$	$t_{0.01}$	$t_{0.005}$	$t_{0.001}$	$t_{0.0005}$
1	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.3	636.6
2	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.33	31.60
3	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.22	12.92
4	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291

Table 6: Probabilities under the t-Distribution



	Right-Tail Probability								
df	0.995	0.99	0.975	0.95	0.9	0.1	0.05	0.025	0.01
1	0.00	0.00	0.00	0.00	0.02	2.71	3.84	5.02	6.63
2	0.01	0.02	0.05	0.10	0.21	4.61	5.99	7.38	9.21
3	0.07	0.11	0.22	0.35	0.58	6.25	7.81	9.35	11.34
4	0.21	0.30	0.48	0.71	1.06	7.78	9.49	11.14	13.28
5	0.41	0.55	0.83	1.15	1.61	9.24	11.07	12.83	15.09
6	0.68	0.87	1.24	1.64	2.20	10.64	12.59	14.45	16.81
7	0.99	1.24	1.69	2.17	2.83	12.02	14.07	16.01	18.48
8	1.34	1.65	2.18	2.73	3.49	13.36	15.51	17.53	20.09
9	1.73	2.09	2.70	3.33	4.17	14.68	16.92	19.02	21.67
10	2.16	2.56	3.25	3.94	4.87	15.99	18.31	20.48	23.21
11	2.60	3.05	3.82	4.57	5.58	17.28	19.68	21.92	24.72
12	3.07	3.57	4.40	5.23	6.30	18.55	21.03	23.34	26.22
13	3.57	4.11	5.01	5.89	7.04	19.81	22.36	24.74	27.69
14	4.07	4.66	5.63	6.57	7.79	21.06	23.68	26.12	29.14
15	4.60	5.23	6.26	7.26	8.55	22.31	25.00	27.49	30.58
16	5.14	5.81	6.91	7.96	9.31	23.54	26.30	28.85	32.00
17	5.70	6.41	7.56	8.67	10.09	24.77	27.59	30.19	33.41
18	6.26	7.01	8.23	9.39	10.86	25.99	28.87	31.53	34.81
19	6.84	7.63	8.91	10.12	11.65	27.20	30.14	32.85	36.19
20	7.43	8.26	9.59	10.85	12.44	28.41	31.41	34.17	37.57
22	8.64	9.54	10.98	12.34	14.04	30.81	33.92	36.78	40.29
24	9.89	10.86	12.40	13.85	15.66	33.20	36.42	39.36	42.98
26	11.16	12.20	13.84	15.38	17.29	35.56	38.89	41.92	45.64
28	12.46	13.56	15.31	16.93	18.94	37.92	41.34	44.46	48.28
30	13.79	14.95	16.79	18.49	20.60	40.26	43.77	46.98	50.89
40	20.71	22.16	24.43	26.51	29.05	51.81	55.76	59.34	63.69
50	27.99	29.71	32.36	34.76	37.69	63.17	67.50	71.42	76.15
60	35.53	37.48	40.48	43.19	46.46	74.40	79.08	83.30	88.38
70	43.28	45.44	48.76	51.74	55.33	85.53	90.53	95.02	100.43
80	51.17	53.54	57.15	60.39	64.28	96.58	101.88	106.63	112.33
90	59.20	61.75	65.65	69.13	73.29	107.57	113.15	118.14	124.12
100	67.33	70.06	74.22	77.93	82.36	118.50	124.34	129.56	135.81

Table 7: Probabilities under the χ^2-Distribution

Part IV: Formulae and Notation

19 | Formulae

Statistic	Formula
Crosstabulations	
χ^2-Test Statistic	$\chi^2 = \sum \frac{(f_o - f_e)^2}{f_e}$
Two Sample Tests	
Standard Error of Difference	$se = \sqrt{(se_1)^2 + (se_2)^2}$
Means	
Standard Error of Difference	$se = \sqrt{\frac{se_1^2}{n_1} + \frac{se_2^2}{n_2}}$
Standard Error (pooled)	$se_0 = \sqrt{s_0^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$
Variance (pooled)	$s_0^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$
Proportions	
Standard Error	$se = \sqrt{\frac{\hat{\pi}(1 - \hat{\pi})}{n}}$
Standard Error of Difference	$se = \sqrt{\frac{\hat{\pi}_1(1 - \hat{\pi}_1)}{n_1} + \frac{\hat{\pi}_2(1 - \hat{\pi}_2)}{n_2}}$
Standard Error (pooled)	$se_0 = \sqrt{\bar{\pi}(1 - \bar{\pi})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$
Regression	
Adjusted R^2	$\bar{R}^2 = 1 - \frac{\sum \hat{\epsilon}_i^2 / (n - k - 1)}{\sum (y_i - \bar{y})^2 / (n - 1)}$
Coefficient of Determination	$R^2 = \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS} = 1 - \frac{\sum \hat{\epsilon}_i^2}{\sum (y_i - \bar{y})^2}$
Estimated Variance	$\hat{\sigma}^2 = \frac{\sum \hat{\epsilon}_i^2}{n - p}$
Estimator of β_0	$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$; $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{y})$
Estimator of β_1	$\hat{\beta}_1 = \frac{\sum_{i=1}^{N} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{N} (x_i - \bar{x})^2}$
Explained Sum of Squares	$\sum (\hat{y}_i - \bar{y})^2$
Residual Sum of Squares	$\sum \hat{\epsilon}_i^2 = \sum (y_i - \hat{y}_i)^2$
Estimated Standard Error of $\hat{\beta}_0$	$\hat{se}(\hat{\beta}_0) = \sqrt{\frac{\sum x_i^2}{n \sum (x_i - \bar{x})^2}}\, \hat{\sigma}$; $\hat{\text{var}}(\hat{\boldsymbol{\beta}}) = \hat{\sigma}^2 (\boldsymbol{X}'\boldsymbol{X})^{-1}$
Estimated Standard Error of $\hat{\beta}_1$	$\hat{se}(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{\sum (x_i - \bar{x})^2}}$
Total Sum of Squares	$\sum (y_i - \bar{y})^2$

Table 8: Formulae for PO12Q

19.1 Matrices

$$\boldsymbol{X'X} = \begin{bmatrix} 1 & 1 & \dots & 1 \\ x_1 & x_2 & \dots & x_n \end{bmatrix} \times \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} = \begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix}$$

$$(\boldsymbol{X'X})^{-1} = \frac{1}{n \sum x_i^2 - \sum x_i \sum x_i} \begin{bmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{bmatrix}$$

$$\boldsymbol{X'y} = \begin{bmatrix} 1 & 1 & \dots & 1 \\ x_1 & x_2 & \dots & x_n \end{bmatrix} \times \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \end{bmatrix}$$

Symbol	Explanation
χ^2	Chi-Squared for test of independence
π	Population Proportion
$\hat{\pi}$	Sample Proportion
$\bar{\pi}$	Pooled Proportion
se_0	Standard Error under the Null Hypothesis
β	Regression Coefficient
$\hat{\beta}$	Estimated Regression Coefficient
ϵ	Error Term
$\hat{\epsilon}$	Estimated Error Term / Residual
A^{-1}	Inverse of Matrix A
A'	Transpose of Matrix A
I	Identity Matrix
R^2	R-Squared / Model Fit
σ^2	Mean Squared Error
k	Number of Slope Coefficients
$\bar{\mathrm{R}}^2$	Adjusted R-Squared
log(x)	Logarithm of variable x
α_i	Regression coefficients of secondary regression models
p	Total Number of Regression Coefficients, including the Intercept
VcV	Variance-Covariance Matrix
Ω	Is equal to $\sigma^2 I$, where σ^2 represents the mean squared error
t	Time period in time-series data

Table 9: Notation

Part V: References

List of References

Miller, M., Boix, C., & Rosato, S. (2022). *Boix-Miller-Rosato Dichotomous Coding of Democracy, 1800-2020*. https://doi.org/10.7910/DVN/FENWWR

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